

C813

Healthcare Statistics and Research

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A 250-bed hospital that won a state safety award recorded 6,200 adult and pediatric inpatient service days in June (30 days). It also had 1,180 discharges and 42 newborn births that month. What was the average daily inpatient census?

- A. 82.7%
- B. 24.8
- C. 206.7
- D. 39.3

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A 120-bed rehabilitation hospital with a busy outpatient clinic recorded 3,348 inpatient service days in January (31 days). Its outpatient clinic had 2,900 visits that month. What was the inpatient bed occupancy rate?

- A. 108
- B. 90%
- C. 111%
- D. 168%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A 200-bed hospital had an 85% occupancy rate in a 30-day month. How many inpatient service days did it record?

- A. 7,059
- B. 6,000
- C. 170
- D. 5,100

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A community hospital whose new CEO came from a large academic center discharged 400 adult patients in March. Those patients had a combined 1,800 days of stay. The hospital's inpatient service days for March were 1,900. What was the average length of stay?

- A. 0.22 days
- B. 4.5 days
- C. 61.3
- D. 4.75 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A unit's average length of stay was 5.2 days across 250 discharges. What was the total length of stay in discharge days?

- A. 5.2 days
- B. 255.2 days
- C. 1,300 days
- D. 48.1 days

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A hospital that recently joined a national quality collaborative had 1,000 discharges, including deaths, in a quarter. There were 20 inpatient deaths, 5 of them within 48 hours of admission. The hospital also had 3 fetal deaths. What is the gross death rate?

- A. 2.0%
- B. 1.5%
- C. 2.3%
- D. 50

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A clinic manager with a background in finance reviews wait times, in minutes, for 7 patients on a Tuesday: 8, 10, 11, 12, 14, 15, and 95. The 95-minute wait happened during a fire drill. Which measure best describes a typical wait, and what is its value?

- A. The mean after deleting the fire-drill wait, about 11.7 minutes.
- B. The mean, about 23.6 minutes.
- C. The range, 87 minutes.
- D. The median, 12 minutes.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A set of lengths of stay is roughly symmetric with no outliers. Which statement is true?

- A. The mean and median are about equal, and the mean is a fine summary.
- B. The median must always be used for length-of-stay data.
- C. The mean will be much higher than the median.
- D. The mode must equal the range.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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9.

A patient-satisfaction survey at an outpatient center, mailed in both English and Spanish, asks patients to rate pain relief as none, mild, moderate, or complete. The survey had a 38% response rate. What level of measurement is the pain-relief variable?

- A. Interval.
- B. Ordinal.
- C. Ratio.
- D. Nominal.

Target word _____	The one fact that decides _____	Answer ___	[] sure [] unsure [] guess
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10.

An HIM analyst at a system with 14 hospitals finds that units with more nursing informatics training hours have lower medication error rates ($r = -0.62$), based on a retrospective review of records. The chief nursing officer wants to announce that training causes fewer errors. What is the best response?

- A.** The data show an association, not cause; a controlled design would be needed to claim causation.
- B.** The negative r means training increases errors.
- C.** No conclusion is possible because the data came from 14 hospitals.
- D.** An r of -0.62 is strong, so it proves causation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A research team at a hospital that recently earned Magnet designation compares fall rates before and after a new bed-alarm protocol. With $\alpha = 0.05$, the test gives $p = 0.03$. The study included 2 medical units and ran for 6 months. What is the correct conclusion?

- A.** The protocol is proven to prevent all falls.
- B.** There is a 3% chance the protocol works.
- C.** Reject the null hypothesis; the difference is statistically significant at 0.05.
- D.** Fail to reject the null, because 0.03 is less than 0.05.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A study concludes that a new discharge-teaching program has no effect on readmissions, but in truth the program does reduce readmissions. Which error occurred?

- A.** Sampling bias.
- B.** No error, because p above 0.05 proves no effect.
- C.** Type I error.
- D.** Type II error.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A 250-bed hospital ~~that won a state safety award~~ recorded 6,200 adult and pediatric inpatient service days in June (30 days). ~~It also had 1,180 discharges and 42 newborn births that month.~~ What was the average daily inpatient census?

Target word: average daily census **Decides:** 6,200 service days / 30 days.

A	Adjacent category	That is the occupancy rate ($206.7 / 250$), not the census.
B	Confusion set	Divides service days by beds, not days.
C	Correct	$6,200 / 30 = 206.7$ patients per day.
D	Camouflage	Uses discharges ($1,180 / 30$), which do not measure census.

2. Answer B Concept 2

A 120-bed rehabilitation hospital ~~with a busy outpatient clinic~~ recorded 3,348 inpatient service days in January (31 days). ~~Its outpatient clinic had 2,900 visits that month.~~ What was the inpatient bed occupancy rate?

Target word: occupancy rate **Decides:** Bed-days available = $120 \times 31 = 3,720$.

A	Adjacent category	That is the average daily census ($3,348 / 31$), not a percent of beds.
B	Correct	$3,348 / 3,720 = 0.90 = 90\%$.
C	Inverted operation	Divides available bed-days by service days.
D	Camouflage	Adds outpatient visits, which do not use inpatient beds.

3. Answer D Concept 2 · transfer

A 200-bed hospital had an 85% occupancy rate in a 30-day month. How many inpatient service days did it record?

Target word: solve for service days **Decides:** Service days = beds $\times$ days $\times$ rate.

A	Inverted operation	Divides bed-days by 0.85 instead of multiplying.
B	Confusion set	That is available bed-days, not days used.
C	Dropped element	That is the average daily census; drops the 30 days.
D	Correct	$200 \times 30 \times 0.85 = 5,100$.

4. Answer B Concept 3

A community hospital ~~whose new CEO came from a large academic center~~ discharged 400 adult patients in March. Those patients had a combined 1,800 days of stay. ~~The hospital's inpatient service days for March were 1,900.~~ What was the average length of stay?

Target word: ALOS **Decides:** Discharge days 1,800 / 400 discharges.

A	Inverted operation	Divides discharges by days.
B	Correct	$1,800 / 400 = 4.5$ days.
C	Adjacent category	That is average daily census ($1,900 / 31$).
D	Camouflage	Uses service days (1,900), a census measure, not discharge days.

5. Answer C Concept 3 • transfer

A unit's average length of stay was 5.2 days across 250 discharges. What was the total length of stay in discharge days?

Target word: solve for total days **Decides:** Total = ALOS × discharges.

A	Wrong stem number	That is the average given.
B	Confusion set	Adds instead of multiplies.
C	Correct	$5.2 \times 250 = 1,300$.
D	Inverted operation	Divides discharges by ALOS.

6. Answer A Concept 4

A hospital ~~that recently joined a national quality collaborative~~ had 1,000 discharges, including deaths, in a quarter. There were 20 inpatient deaths, 5 of them within 48 hours of admission. ~~The hospital also had 3 fetal deaths.~~ What is the gross death rate?

Target word: gross death rate **Decides:** All inpatient deaths / all discharges.

A	Correct	$20 / 1,000 = 2.0\%$.
B	Confusion set	Excludes the 5 early deaths, which is the net death rate.
C	Camouflage	Adds fetal deaths, which are excluded.
D	Inverted operation	Divides discharges by deaths.

7. Answer D Concept 5

A clinic manager ~~with a background in finance~~ reviews wait times, in minutes, for 7 patients on a Tuesday: 8, 10, 11, 12, 14, 15, and 95. ~~The 95-minute wait happened during a fire drill.~~ Which measure best describes a typical wait, and what is its value?

Target word: best center measure **Decides:** One extreme value skews the data.

A	Camouflage	Dropping real data because of its cause is not a valid fix; use the median.
B	Confusion set	$165 / 7 \approx 23.6$, pulled up by the outlier.
C	Adjacent category	The range measures spread, not a typical value.
D	Correct	The middle of seven ordered values is 12, and the median resists the outlier.

8. Answer A Concept 5 • transfer

A set of lengths of stay is roughly symmetric with no outliers. Which statement is true?

Target word: mean vs. median — symmetric case **Decides:** No skew.

A	Correct	In symmetric data the two measures agree.
B	Absolute language	The median is preferred only when data are skewed.
C	Inverted operation	That describes right-skewed data.
D	Confusion set	Mixes a center measure with a spread measure.

9. Answer B Concept 6

A patient-satisfaction survey at an outpatient center, ~~mailed in both English and Spanish~~, asks patients to rate pain relief as none, mild, moderate, or complete. ~~The survey had a 38% response rate.~~ What level of measurement is the pain-relief variable?

Target word: level of measurement **Decides:** Ordered categories without equal spacing.

A	Confusion set	Assumes equal distances between categories.
B	Correct	The categories have a clear order but no measured distance between them.
C	Camouflage	The response rate is a ratio measure; the pain variable is not.
D	Dropped element	Ignores the ranking built into the labels.

10. Answer A Concept 7

An HIM analyst ~~at a system with 14 hospitals~~ finds that units with more nursing informatics training hours have lower medication error rates ($r = -0.62$), based on a retrospective review of records. The chief nursing officer wants to announce that training causes fewer errors. What is the best response?

Target word: what a correlation supports **Decides:** Observational association only.

A	Correct	Correlation does not rule out other explanations.
B	Inverted operation	Negative r means as training rises, errors fall.
C	Camouflage	The number of hospitals is not the limitation; the design is.
D	Absolute language	Strength of association never proves cause.

11. Answer C Concept 8

A research team at a hospital ~~that recently earned Magnet designation~~ compares fall rates before and after a new bed-alarm protocol. With $\alpha = 0.05$, the test gives $p = 0.03$. ~~The study included 2 medical units and ran for 6 months.~~ What is the correct conclusion?

Target word: decision rule **Decides:** $p(0.03) < \alpha(0.05)$.

A	Absolute language	Significance shows a difference, not proof or elimination of falls.
B	Confusion set	p is the chance of data this extreme if the null is true, not the chance the protocol works.
C	Correct	A p -value below α leads to rejecting the null.
D	Inverted operation	Reverses the decision rule.

12. Answer D Concept 8 · transfer

A study concludes that a new discharge-teaching program has no effect on readmissions, but in truth the program does reduce readmissions. Which error occurred?

Target word: name the error — false negative **Decides:** Failed to reject a false null.

A	Adjacent category	Bias may cause errors, but the error named here is a decision error.
B	Absolute language	Failing to find significance never proves there is no effect.
C	Inverted operation	Type I is finding an effect that is not there.
D	Correct	Missing a real effect is a Type II (false negative) error.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Average daily inpatient census	Average daily census = total inpatient service days / number of days in the period.	Dividing by beds or using discharges.	Service days and a period length.
2	Inpatient bed occupancy rate	Occupancy = inpatient service days / (bed count × days in period) × 100.	Stopping at the average census or inverting the ratio.	A bed count, service days, and days in the month.
3	Average length of stay	ALOS = total length of stay (discharge days) / number of discharges.	Using inpatient service days (a census measure) instead of discharge days.	Both service days and discharge days appear in the stem.
4	Gross vs. net death rate	Gross death rate = inpatient deaths / discharges (including deaths); the net rate excludes deaths within 48 hours of admission; fetal deaths are excluded from both.	Mixing the net and gross rules or adding fetal deaths.	Deaths under 48 hours and fetal deaths listed separately.
5	Mean vs. median with outliers	The median resists extreme values; the mean is pulled toward outliers. Use the median for skewed data.	Reporting the mean for skewed data, or deleting outliers.	One value far from the rest.
6	Levels of measurement	Nominal = categories without order; ordinal = ordered categories with unequal spacing; interval = equal spacing, no true zero; ratio = equal spacing with a true zero.	Calling ranked categories nominal or interval.	Ordered labels like no ne-mild-moderate-complete.
7	Correlation is not causation	An observed association, even a strong one, does not prove cause; causal claims need a controlled design.	Reading a strong r as proof of cause, or a negative r as a harmful effect.	An r value and a leader who wants to say one thing “causes” another.
8	Hypothesis testing and error types	If $p < \alpha$, reject the null. Type I error = rejecting a true null (false positive); Type II = failing to reject a false null (false negative).	Reading p as the probability the treatment works, or swapping Type I and II.	An α , a p-value, or a “no effect” conclusion that turns out wrong.

Course scope (public description): Healthcare Statistics and Research explores the use of statistical data to support process improvement through health information research. Health information management (HIM) professionals use information systems to gather, analyze, and present data in response to administrative and clinical needs. This course has no prerequisites.